

Validation of the EQ-HWB-9 in a mental health sample and an investigation of modifications to items

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Cate Bailey, Karen Trapani, Jonathan N. Davies, Nicholas T. Dam, Julieta Galante & Tessa Peasgood

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Title: Validation of the EQ-HWB-9 in a mental health sample and an investigation of modifications to items

Authors

Cate Bailey^{1,2} ORCID ID: 0000-0001-5030-430X

Karen Trapani³ ORCID ID: 0000-0002-3257-9147

Jonathan N Davies² ORCID ID: 0000-0001-7838-126X

Nicholas T Van Dam² ORCID ID: 0000-0002-1131-0739

Julieta Galante² ORCID ID: 0000-0002-4108-5341

Tessa Peasgood⁴ ORCID ID: 0000-0001-8024-7801

¹Melbourne Health Economics, School of Population and Global Health, The University of Melbourne, Carlton, Victoria, Australia

²Contemplative Studies Centre, Melbourne School of Psychological Sciences, The University of Melbourne, Carlton, Victoria, Australia

³Centre for Health Policy, School of Population and Global Health, The University of Melbourne, Carlton, Victoria, Australia

⁴Sheffield Centre for Health and Related Research, School of Medicine and Population Health, University of Sheffield, Sheffield, UK

Corresponding author:

Dr Cate Bailey,

207 Bouverie St, Carlton, 3035, Victoria, Australia.

cate.bailey@unimelb.edu.au

Declarations

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Competing interests Dr Peasgood is a member of the EuroQol group, the EQ-HWB Working Group, and was involved with the development of the EQ-HWB. Dr

Bailey is a member of the EQ-HWB special interest group. The authors have no competing interests to declare that are relevant to the content of this article.

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Abstract

Background

The EQ Health and Wellbeing (EQ-HWB) is a new, generic instrument designed to evaluate quality-of-life across health, public health, and social care settings. The short form comprises nine items (EQ-HWB-9) and validation across diverse populations and contexts is required. We aimed to investigate the validity of the EQ-HWB-9 in an international sample of adults experiencing poor mental health who downloaded a meditation app. We further examined the impact of four item-level modifications on psychometric performance, including investigating an ordering effect for the 'activities' item (as noted in prior studies) and three minor changes to response options.

Methods

The current study was embedded in a larger trial examining engagement with meditation via a free, downloadable app. Participants were randomised to complete the original (2022) and modified (2024) experimental version of the EQ-HWB-9. Psychometric evaluation included analyses of item distribution, known group and convergent validity, and responsiveness to change.

Results

There were no differences in demographic characteristics between the EQ-HWB original and modified versions at baseline (n=865) or follow-up (n=130). All psychometric tests supported the validity of the EQ-HWB-9 in this population. We found an ordering effect for the activities item, where the modified activities item had greater level of difficulty and a wider distribution over response options when asked before the mobility item, rather than after. There were no observable differences between versions for the other modifications.

Conclusions

These findings add to growing literature supporting the EQ-HWB-9 as a suitable instrument for measuring quality-of-life across a range of settings. When the 'activities' item was presented first, as in the modified version, participants appeared to interpret the item more broadly, in line with the developers' intentions. Accordingly, our results support changing the item order of the first two items. Other modifications had little impact on outcomes, suggesting that further qualitative research will be required to inform decisions about their inclusion. Results from this study provide support towards finalisation of the instrument. The development of country-specific value-sets is now critical to support its application in economic evaluation

Key words

EQ-HWB; EuroQol Health and Wellbeing; Quality-of-life; instrument validation; psychometrics; mental health

Introduction

The EQ Health and Wellbeing (EQ-HWB) is a recently developed generic instrument designed to evaluate quality-of-life across health, public health, and social care settings(1). Two versions exist: a 25-item full instrument (EQ-HWB) and a 9-item short form (EQ-HWB-9). Recent studies have found that the EQ-HWB-9 has robust psychometric outcomes in caregiver populations(2–7) and in general populations in China, UK and Australia(7–10). Psychometric testing has included tests of known-group and convergent validity(2,3,6,7,9) and responsiveness to change(2,11). A pilot UK value set for the EQ-HWB-9 ('experimental', 2022, v1.1) has been published(12), enabling calculation of quality-adjusted life years (QALYs). Further testing of the instruments in a range of populations is required, particularly in those populations for which the instrument was intended, such as caregivers, social care users, and in mental health (13).

Mental health disorders are a leading cause of global disease burden, affecting individuals' functioning, relationships, and quality-of-life across the life course(14). Increasing rates of depression, anxiety, and stress-related conditions, combined with limited access to timely and effective mental healthcare, have led to growing interest in digital health interventions(15), with recent research finding that mindfulness apps have small but significant effects in reducing anxiety and depression(16). Prior research shows that meditation app users tend to have poorer mental health status than the general population; in a recent study of 886 meditators in the United States, nearly one-quarter (24%) met criteria for a moderate or severe mental disorder(17).

Understanding the impact of meditation apps is important not only for clinical practice but also for informing decisions about resource allocation. Economic evaluation of these interventions requires robust, preference-based measures of health and wellbeing that can capture relevant domains of impact(18). Digital mental health interventions often target psychosocial outcomes that align with EQ-HWB-9 domains, particularly anxiety, low mood, fatigue, and reduced sense of control. Therefore, evaluating the psychometric performance of the EQ-HWB-9 in users of meditation apps can inform its suitability for economic evaluations of digital mental health programs.

The EQ-HWB instruments are in the final stages before being released for general use, with finalisation of the short-form being a priority for the developers due to its relevance for economic evaluation, noting that the long-form would also be highly relevant in this population. During this time, a range of modifications are being considered such as changes to item order, and minor wording changes to items and response options. In our previous qualitative work, we observed context effects, in that that some participants perceived the activities item as a physical item when presented after mobility; however, when queried, participants stated much of their difficulty with day-to-day items was caused by psycho-social reasons such as exhaustion and caring responsibilities(2). When participants were asked about the order of items, most participants supported changing the item order to present activities first(4). We were therefore interested in whether participants perceived the activities item more broadly if it was presented prior to the mobility item. Testing the proposed modifications to the EQ-HWB-9 is now a priority for instrument developers.

The aim of this study was to use psychometric testing to assess whether the EQ-HWB-9 was a suitable quality-of-life measure in a sample of adults who downloaded a mindfulness meditation app.. We examined response distributions, mean differences and correlations between items, known-group validity at the sum-score and index-score level, convergent validity with measures of mental health symptoms and wellbeing, and responsiveness to change in this population. Because of the large sample size expected for the study, we were able to test a set of modifications being considered for the EQ-HWB-9 by randomising participants to the original or modified version and investigate any differences in outcomes whilst maintaining sufficient power for the psychometric analyses. Hence, we further aimed to compare the EQ-HWB-9 ('experimental', 2022, v1.1) version to the EQ-HWB-9 ('experimental modified', 2024, v1.2) version of the EQ-HWB-9 to explore the effect of the modifications on the instruments' psychometric properties in this sample.

2.0 Method

2.1 Study design and procedure

On downloading the meditation app, an in-app notification invited users to participate in a research study. This notification then directed participants outside the app to information on the study, a consent form, and a link to the online survey. Data were collected between July and September 2024. Participants completed brief self-report measures in English on day 0 (baseline) and day 28 (post-intervention). Participants needed to be over 18 years of age. The current study was a “piggy-back” study; the main aim of the study was to investigate participant engagement with a meditation app.

2.2 Materials

Data collection included a range of demographic variables (age, gender, educational attainment, family income, employment status and geographic location) and information on mental health and app use (past meditation practice, the use of mindfulness apps and any previous unpleasant experiences of meditation practice, including the functional impact and duration of reported unpleasant experiences, ever or currently receiving professional mental health support). Participants provided information about their combined total family income by selecting an income band from a 5-tier country specific income range. The income range that the participant was presented with was based on their country of residence. We coded this into a common 5-tier income range where the lowest band was mapped to the value “Lowest” and the highest band was mapped to the value “Highest” and the intermediate bands were labelled “2”, “3” and “4”. This allowed a comparison of income band demographics across the cohort.

Data from five instruments was collected. The **EQ-HWB-9** included 9-items covering physical (activities, mobility, pain) and psychosocial items (exhaustion, loneliness, cognition, anxiety, sadness/depression, lack of control) over the last seven days. Each item has five response options. EQ-HWB-9 sum scores were calculated by summing individual item scores. The index-score was calculated by applying a pilot EQ-HWB-9 value-set from the UK(12). The **Kessler Psychological Distress Scale (K10)**(19) is a 10-item questionnaire of psychological distress measured over the previous 30-days. The scale ranges from 1 (none of the time) to 5 (all of the time). K10 sum scores were calculated by summing each of the ten items. Cut points were defined as likely to be well or have a mild disorder (≤ 24) or likely to have a moderate or severe disorder (> 25)(20). The **Short Warwick-Edinburgh Mental Wellbeing Scale (SWEMWBS)**(21) assesses positive facets of

mental health on a 7-item, 5-point scale: 1 (None of the time), 2 (Rarely), 3 (Some of the time), 4 (Often) and 5 (Always), calculated by summing all items. Data were adjusted to produce a metric score(22). We used the cut points for of above 28.1 as high, and below 19.3 as low Cut-points for three groups were defined as low (7-19.3), medium (20-27.0) and high (28.1-35.0) wellbeing (23). The **Perceived Stress Scale** (PSS-4)(24) was used to assess the participant's perceived stress in the past month with four-items and using four response options 0 (Never), 1(Almost never), 2 (Sometimes), 3 (Fairly often), 4 (Very often). Items 2 and 3 are reverse-scored and the items are then summed. Cut-points were defined as no or low stress (<6) compared to higher stress (≥ 6)(25).The **Satisfaction with Life Survey** (1-item) (SWLS1)(26) is an abbreviated measure of the Satisfaction with Life Survey with a single-item asking participants to rate life satisfaction on a 7-item scale ranging from 1 (extremely dissatisfied) to 7 (extremely satisfied).

Self-reported utilisation of professional mental health support was measured through the following question: *"Have you ever received professional mental health support? This could include support from a primary care provider, GP, family medicine doctor, therapist, social worker, psychologist or psychiatrist, online or in person"* (No / Yes - received support in the past / Yes - am receiving support now). "Current mental health support" was coded as participants who were only currently receiving mental health support versus all others, and "Any mental health support" for participants who reported current or previous use versus 'no'.

Sleep was measured using the EQ-HWB long form Item 9, with permission from the developers, *"In the last 7 days, did you have problems with your sleep?"* on a five-point response scale (sleep is not included in the short-form). "Sleep

problems” were defined as yes (sometimes/often/most of the time) and no (none of the time/only occasionally). Unpleasant experiences with meditation were measured using the question: “*Have you ever had any particularly unpleasant experiences (e.g., anxiety, fear, distorted emotions or thoughts, altered sense of self or the world) during your overall meditation practice?*”) (yes/no). Participants who responded affirmative were asked to respond to an additional statement “*My meditation-related challenging, difficult or distressing experiences impaired my ability to function*” by selecting from a 4-item answer scale (not at all/somewhat/moderately/severely). We coded a binary variable “*Functional impairment from meditation*” for participants who responded with somewhat, moderately or severely, compared to those who answered not at all and who had indicated no unpleasant experiences in the previous question.

2.3 Missing data

The study dataset included baseline survey responses from 1021 participants. We excluded survey responses if participants had not completed the five instruments (156 records) for a final baseline sample of 865. We had 130 participants with EQ-HWB-9 data at follow-up. Missing data analysis suggested that participants who did not complete data collection were more likely to be male, and significantly but with small percentage differences for education (slightly lower) and occupation (slightly less likely to work full time), and slightly younger (not significant) (Table S8).

2.4 Differences between original and modified versions of the EQ-HWB-9

Participants were randomised to complete either the EQ-HWB-9 ('experimental', 2022, v1.1) or the EQ-HWB-9 ('experimental modified', 2024, v1.2) version of the instrument . We tested the following modifications: 1) swapping the

order of the first two items (in the modified version activity now preceded mobility); 2) adjusting the third response option of the difficulty response scale (items 1 and 2) from “**Some** difficulty” to “**Moderate** difficulty”; 3) Adjusting the second response option of the frequency response scale (items 3-8) from “**Only occasionally**” to “**A little of the time**”; and 4) adjusting the wording of Item 8 from “In the last 7 days, did you feel you **had no** control over your day-to-day life? (e.g. had no choice **to do things or have things done for you as you liked and when you wanted**)” to “In the last 7 days, did you feel you **lacked** control over your day-to-day life? (e.g. had no choice **over what you did or how things were done for you**)”.

2.5 Statistical Analysis

Analyses were conducted separately for groups completing the original and the modified versions, except when being directly compared. Demographic characteristics were compared between participants in the two groups using chi-squared and *t*-tests. Response distributions were calculated for item scores using frequency and percentages for the baseline and follow-up surveys. We compared mean scores of the items across the two versions using *t*-tests. We correlated EQ-HWB-9 items to investigate differences between versions. We used the CORTESTI module in STATA to test the equality of item-level correlation coefficients between original and modified versions(27).

Known group validity analysis was conducted at the sum-score and index-score levels using *t*-tests (binary outcomes) and one-way ANOVAs (3 or 4 groups). EQ-HWB-9 sum-scores and preference-weighted scores were treated as continuous measures. Effect sizes were measured using Cohen’s *d* (0.2 small, >0.5 moderate, <0.8 large(28)). We hypothesised that lower quality-of-life (higher EQ-HWB-9 sum-

scores, lower EQ-HWB-9 index-scores) would be associated with poorer mental health status (K10)(2,8), lower well-being (SWEMWBS)(11,29), higher psychological stress (PSS-4) and lower satisfaction with life (SWLS1) (both not previously compared to EQ-HWB), and sleep problems(8). We hypothesised that participants receiving current or any professional mental health support would have lower quality-of-life than participants not receiving professional mental health support, on the basis that they were likely to have had mental health issues in order to seek support (acknowledging that this hypothesis could be reversed if mental health support was highly successful, and those not receiving support had needed it). We hypothesised that unpleasant experiences and functional impairment associated with meditation practice would be associated with lower quality-of-life, although committed meditators may be more likely to have experienced and unpleasant effect.

We used Spearman's rank-order correlation (coefficient) for ordinal data to compare the EQ-HWB-9 and K10, SWEMWBS and PSS4 items and total scores. We defined correlation strength as 0 - 0.09 no correlation, 0.1-0.29 weak, 0.3-0.49 moderate, and ≥ 0.5 strong(28). Hypotheses were determined *a priori* with consensus (TP and CB) for items theoretically likely to be correlated.

For the responsiveness to change analyses we calculated the EQ-HWB-9 change in sum-score between baseline and follow-up (follow-up minus baseline) for the 130 cases for which we had follow-up data. Change scores were calculated for the K10, SWEMWBS and PSS4 instruments in the same way, and then recoded into three groups for 'improved', 'no change', and 'worsened' scores based on the distribution of the scores, as shown in Table S1. For the EQ-HWB-9 to be seen as

responsive to change, we would expect that deteriorating scores on the K10, SWEMBWS, PSS4 and SWLS1 would be associated with deteriorating quality-of-life scores and vice versa. Unfortunately, the versions were inadvertently “re-randomised” by the data collection team at follow-up. The re-randomisation created four groupings, such that we had half the number of participants we could track across time with the same randomisation: 1) experimental version at baseline and follow-up, 2) experimental version at baseline and modified version at follow-up, 3) modified version at baseline and experimental version at follow-up, 4) modified version at baseline and follow-up. In the manuscript, we have presented data for the EQ-HWB sum-scores combining all four groups (due to the low sample-size) and presented the subgroup analysis (by the four groups) in supplementary files. We included correlations between EQ-HWB change scores and change scores from the other outcome measures.

Results

3.1 Comparison of baseline differences between original and modified samples

The baseline sample of 865 participants was predominantly below 39 years of age (79%), 55% female, well educated (only 3% did not complete school), and employed full time (41%) (Tables S2a/b). As expected, there were high rates of psychological distress, with 45.2% of participants likely to have moderate to severe disorders on the K10. Participants were from Europe (44%) (Table S2c for full list), United States (29%), Australia (9%), United Kingdom (8%) Canada (8%) and New Zealand (2%). There were no statistically significant differences in baseline demographic characteristics or mean differences in outcome measures (K10, SWEMBWS, PSS4 and SWLS1) between participants randomised to the original

version of the EQ-HWB-9 (n=450) compared to the modified version (n=415) (Tables S2a/b). The follow-up response rate was low at 130 participants (15.0%).

3.2 EQ-HWB-9 item distributions for original and modified versions

Figures 1 and 2 show the distributions of the original and modified versions of the instrument. Numbers and percentages with percentage difference across the response options are displayed in Table S3.

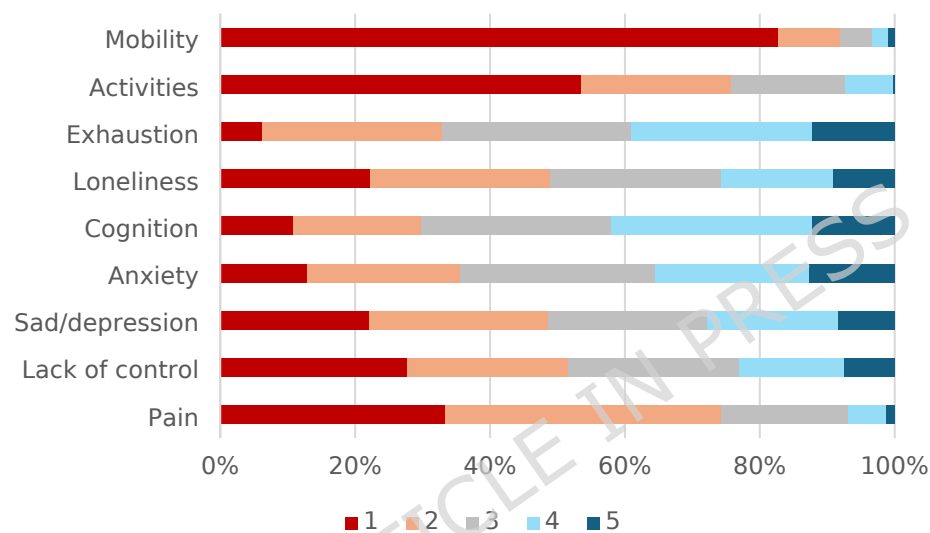


Figure 1 - distribution of EQ-HWB-9 scores for the original version at baseline (n=450)

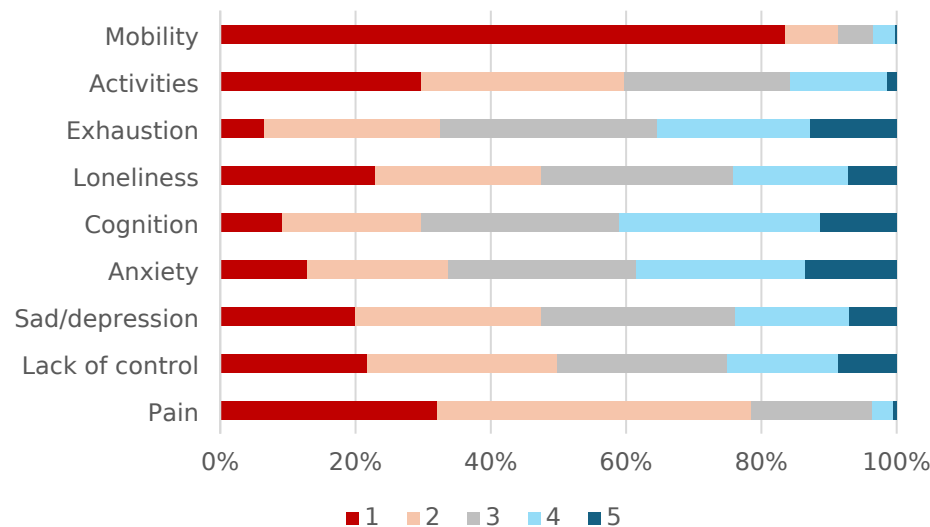


Figure 2 - distribution of EQ-HWB-9 scores for the modified version at baseline (includes all modifications) (n=415)

Note: x-axis numbers correspond to response options, which are different across items

3.3 Comparing items; mean scores comparisons and correlations

The mean scores of the original version of the activities item (1.79, SD=0.98) were statistically significantly lower than the mean scores for the modified version (2.28, SD=1.08) (mean difference = .50, $t(df) = 7.06(881)$, $p < .001$, Cohen's $d = .48$), suggesting an order effect. There were no statistically significant differences between the original and modified versions for any other items (Table S4). Mean results are displayed in Figure 3.

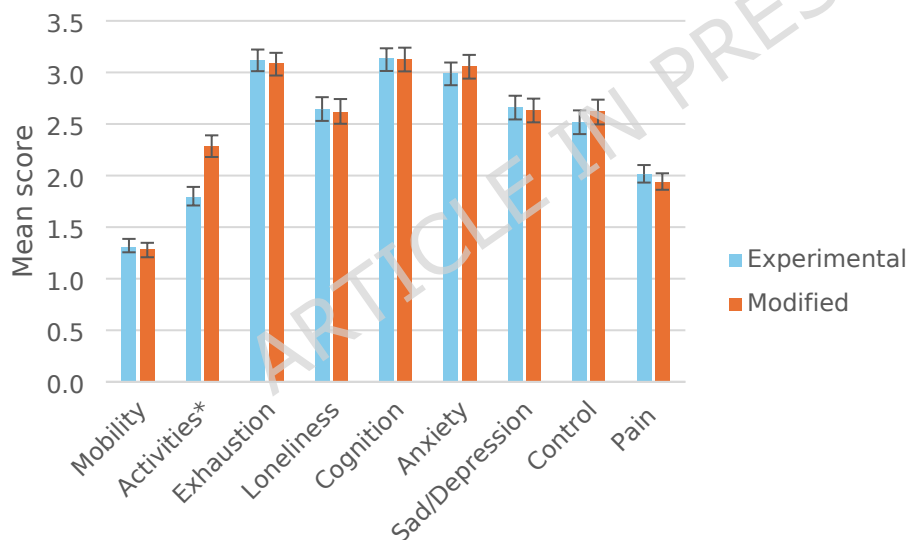


Figure 3 - mean scores of EQ-HWB-9 scores for original and modified versions with 95% confidence intervals

* $p < .001$

We compared correlations between EQ-HWB-9 items for the original and modified versions and highlighted any correlation pairs, as shown in Table 1. This analysis suggests a pattern of differences between original and modified versions of the 'activities' item with other items. For the modified version of the instrument,

the ‘activities’ item had lower correlations with mobility and higher correlations with exhaustion, cognition and sad/depression items.

Table 1 - Correlations between EQ-HWB-9 items at baseline

		Mobility	Activities	Exhaustion	Loneliness	Cognition	Anxiety
Mobility	Original	1.000***					
	Modified	1.000***					
	$p^{\wedge}$ (diff)	n/a					
Activities	Original	.448***	1.000***				
	Modified	.305***	1.000***				
	$p^{\wedge}$ (diff)	.014 (.143)	n/a				
Exhaustion	Original	.270***	.435***	1.000***			
	Modified	.308***	.566***	1.000***			
	$p^{\wedge}$ (diff)	.544 (-.038)	.010 (-.131)	n/a			
Loneliness	Original	.156***	.397***	.398***	1.000***		
	Modified	.167***	.416***	.407***	1.000***		
	$p^{\wedge}$ (diff)	.879 (-.011)	.739 (-.019)	.875 (-.009)	n/a		
Cognition	Original	.191***	.416***	.492***	.484***	1.000***	
	Modified	.123*	.527***	.512***	.403***	1.000***	
	$p^{\wedge}$ (diff)	.307 (.068)	.036 (-.111)	.695 (-.020)	.139 (.081)	n/a	
Anxiety	Original	.187***	.329***	.450***	.503***	.480***	1.000***
	Modified	.128**	.416***	.465***	.412***	.546***	1.000***
	$p^{\wedge}$ (diff)	.139 (.059)	.139 (-.087)	.781 (-.015)	.091 (.091)	.189 (.077)	n/a
Sad/depression	Original	.193***	.400***	.457***	.635***	.436***	.660***
	Modified	.233***	.515***	.498***	.631***	.468***	.579***
	$p^{\wedge}$ (diff)	.540 (-.040)	.033 (-.115)	.437 (-.041)	.922 (.004)	.556 (-.032)	.054 (.081)
Lack of control	Original	.195***	.411***	.406***	.423***	.485***	.470***
	Modified	.184***	.480***	.470***	.440***	.512***	.483***
	$p^{\wedge}$ (diff)	.207 (.011)	.207 (-.069)	.246 (-.064)	.760 (-.017)	.600 (.027)	.806 (-.013)
Pain	Original	.339***	.305***	.341***	.211***	.189***	.185***
	Modified	.292***	.356***	.356***	.165***	.304***	.241***
	$p^{\wedge}$ (diff)	.402 (.047)	.402 (-.051)	.803 (-.015)	.485 (.056)	.073 (-.115)	.390 (-.056)

$\wedge$ p -value for comparison between correlation coefficients

Correlations highlighted in light blue if the p -value was significantly different at $<.05$; p -values highlighted in mid-blue if less than $<.05$. Differences between correlations are in brackets beside the p -values.

* = $p<.05$, ** = $p<.01$, *** = $p<.001$,

3.4 Known-group analyses

We conducted known-group analysis using *t*-tests across a range of variables, shown in full in Table S5a for the sum-score, and Table S5b for the index-scores. In the original version, all tests were significant in hypothesized directions, with *p*-values less than .001 for sum-score analyses, except current mental health support, which was still significant at less than .01. Cohen's *d* scores are summarised in Table 2 to investigate differences between original and modified versions, which appear to be similar. There were no apparent differences between the original and modified versions of the instrument in the known-group analyses by 3 and 4 groups, as shown in Table S5c.

Table 2 - Comparisons of Cohen's *d* scores between the original and modified versions, for sum-score and index-scores

Known groups	Sum-scores (<i>d</i>)			Index-scores (<i>d</i>)		
	Original	Modified	Difference	Original	Modified	Difference
Psychological Distress (K10)	-2.00	-1.93	0.07	1.82	1.77	0.05
Low mental wellbeing (SWEMWBS)	-1.44	-1.42	0.02	1.28	1.33	-0.05
Psychological Stress (PSS4)	-1.43	-1.41	0.02	1.17	1.24	-0.07
Satisfied with life (SWLS1)	1.02	1.06	0.04	-0.97	-0.99	0.02
Current mental health support [*]	-0.32	-0.46	-0.14	0.29	0.49	-0.20
Any mental health support ^{**}	-0.43	-0.55	-0.12	0.38	0.54	-0.16
Sleep problems	-0.96	-1.02	-0.06	0.87	1.00	-0.13
Unpleasant experiences with meditation	-0.74	-0.48	0.26	0.71	0.51	0.20
Functional impairment from meditation	-0.87	-0.67	0.20	0.85	0.74	0.11

*current vs past or none; **current and past vs none

Cohen's *d*: 0.2 = small, >0.5 = moderate, <0.8 = large(28)

3.5 Convergent validity analysis

All hypothesised correlations between EQ-HWB-9 and the K10 (Table S6a), the SWEMWBS (Table S6b), and the PSS4 (Table S6c) were above the threshold of

a moderate correlation (over 0.3) for both the original and modified versions; hence we found no differences between instrument versions in this regards.

3.6 Responsiveness to change

Responsiveness to change analyses for the n=124-130 participants with follow-up EQ-HWB-9 data against the K10, SWEMWBS, PSS4 and SWLS1 were conducted. Original and modified combined results are shown in Table 3. All tests conformed to hypothesised directions. Reduced psychological distress (K10), reduced wellbeing (SWEMBWS) and reduced stress (PSS4) were all associated with improved quality-of-life (EQ-HWB-9). Correlations between the EQ-HWB change sum-scores and the change scores for other outcome measures were .551 for the K10, -.467 for the SWEMWEBS, .452 for the PSS4, and -.407 for the SWLS1. Subgroup results for the original and modified versions are contained in in Tables S7a/b. Due to the small sample sizes across the groups, some tests lacked statistical significance, and comparisons of the effect sizes between the two versions of the instrument could not be conducted.

Table 3 - Responsiveness to change for the EQ-HWB sum-score (combined original and modified versions)

	#	Mean	SD	F(df)	p-value
K10				16.15 (2,124)	<.001
Reduced mental distress	49	-2.59	4.12		
No change	42	-0.79	3.76		
Increased mental distress	36	2.61	4.72		
Total	127	-0.52	4.66		
SWEMWBS				15.45 (2,122)	<.001
Improved wellbeing	34	2.35	4.75		
No change	43	-0.14	3.54		
Reduced wellbeing	48	-2.88	4.41		
Total	125	-0.51	4.7		

PSS4				11.39 (2,127)	<.001
Reduced stress	36	-3.39	4.57		
No change	55	0.42	3.75		
Improved stress	39	0.87	4.74		
Total	130	-0.50	4.63		
SWLS1				15.13 (2,121)	<.001
Improved satisfaction	17	-4.00	3.61		
No change	101	-0.39	4.24		
Reduced satisfaction	6	7.00	6.07		
Total	124	-0.52	4.72		

DISCUSSION

This study examined the validity of the EQ-HWB-9 for use in an adult sample with higher mental health concerns than the general population who downloaded a meditation app. Our assumption that the population would have higher mental health problems was confirmed, with 45% of participants classified as likely to have a moderate to severe disorder compared to 14% in the Australian population in 2022(30). Overall, item distribution for both versions was similar to previous studies(2-5): participants reported higher mean scores for the items *feeling exhausted*, *cognition* and *anxiety* and lower problems on the 'physical' items (*mobility*, *activities*, *pain*). Both versions discriminated well between groups on the known-groups validity analyses, with large observed between-group effect sizes (<0.8) for psychological distress, low mental wellbeing, stress, satisfaction with life, sleep problems, and meditation-related functional impairment, and small-to-moderate between-group effect sizes for unpleasant experiences with meditation and mental health support. All convergent validity analyses conformed to *a priori* expectations. Sample sizes were low for the responsiveness to change analysis, but all results were in the expected directions. Thus, psychometric testing supports the use of the EQ-HWB-9 in this sample with higher mental health problems.

We then examined whether there were differences in psychometric results between the original and modified versions of the EQ-HWB-9 instrument. No significant baseline or follow-up differences were observed between the randomised original and modified samples across any of the measured variables (age, gender, education, occupation, region, income), indicating that the randomised groups were comparable. Accordingly, any observed variation in the distributions or psychometric performance of the two versions would be expected to occur by chance, unless item modifications exerted a genuine influence on outcomes.

We found substantial evidence of an ordering effect for the first two EQ-HWB-9 items. The modified activities item had significantly higher mean scores and a greater response spread than in the original version, suggesting that when the activities item is presented before the mobility item, participants may interpret ‘activities’ more broadly. This result concords with hypotheses from our prior qualitative work, that participants were thinking of activities as being a physical item when the question came after ‘mobility’ as in the original version, rather than other important aspects that affected their ability to do their day-to-day activities, such as being too tired or depressed, or not being able to think well.

The changes in correlations between the modified activities item and exhaustion, cognition, sad/depression and mobility items compared to correlations in the original version support our hypotheses of a broader interpretation of activities when that item is presented first. There were significant differences in correlations between the original and modified versions of the activities item when compared to mobility, exhaustion, cognition, and sadness/depression, which concords with results from our previous qualitative studies, where participants

referred to aspects of mental health, exhaustion and difficulty with thinking as explanations of why they found day-to-day activities difficult(2,4). This broader interpretation of the activities item was more common if it was asked before the mobility item; whereas, if asked after the mobility item, participants were more likely to interpret the item as referring to physical aspects.

There were no differences in psychometric outcomes for the three other modifications. It is possible that these changes are not sufficiently significant to affect psychometric outcomes in this study. Further quantitative and qualitative research will be needed to assess whether to incorporate these changes in a revised version of the instrument. Other factors such as readability, accessibility, suitability for translations and other cultures may be considered.

Strengths and Limitations

This study builds on previous work that has demonstrated the validity of the EQ-HWB-9 in a variety of settings. The large sample size enabled the randomisation of two versions of the EQ-HWB-9. The main limitation to the study was the follow-up sample size for the responsiveness to change analysis, and the study implementation error which meant that participants were randomised again at follow-up, rather than remaining in their original randomisation groups, which further lowered the sample size for analysis. This part of the analysis, therefore, can only be considered preliminary, with low statistical power. However, longitudinal data for the purposes of validating the EQ-HWB-9 is limited, making this information highly valuable. Significant differences between correlations were set to less than 0.05, and it was not possible to determine whether these changes are practically meaningful or only statistically detectable. We note that mental health problems were self-reported, and that this was not a clinical sample. We

used the UK Pilot value-set for both the EQ-HWB-9 ('experimental', 2022, v1.1) version and the EQ-HWB-9 ('experimental modified', 2024, v1.2) of the instrument and acknowledge that this value-set was developed for the original version and may not be as relevant for the modified version. We did not apply multiple testing correction given that group comparisons and responsiveness were specified *a priori* and our interest was in the magnitude of effect size rather than the binary significance threshold, thus limiting the need for any correction. Correlation comparisons were exploratory and the large number of comparisons increases the risk of Type 1 errors. We acknowledge the limitations of using sum-scores, which gives equal weight to all items(31), whilst also noting that an unweighted approach may be suitable in some circumstances(32). There are currently no value-sets for Australia so we used the UK pilot version(12). Given the large number of countries included, participants may not have completed in the survey in their primary language; however, they downloaded the app in English so we can reasonably assume that they are fluent.

Conclusion

This study adds to the considerable literature evaluating the EQ-HWB-9 in a range of settings. In a large adult population sample of relatively young people, many with mental health concerns, we found that the EQ-HWB-9 had robust psychometric properties. We endorse switching the order of the first two items, as the results here quantify what we found in our previous qualitative studies. Other tested modifications had little effect on psychometric results. This study provides evidence to support the use of the instrument in this population as a generic instrument for measuring quality-of-life that the EQ-HWB Working Group can use to make final decisions on the EQ-HWB-9. The authors recommend that the

instrument becomes available to the wider research community. The next critical step is to develop country-specific value-sets to enable the instrument's use in economic evaluation.

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